# ODE Definitions and Terminology

- An **ODE** relates functions of one variable to their derivatives
  - **Order** is the highest derivative order present
  - **Autonomous** means no explicit dependence on $t$
  - An **Equilibrium Solution** of $\boldsymbol{y}' = \boldsymbol{f}(\boldsymbol{y})$ is constant $\boldsymbol{y} = \boldsymbol{y}_*$ with $\boldsymbol{f}(\boldsymbol{y}_*) = \boldsymbol{0}$
  - An **Initial Condition (IC)** specifies $\boldsymbol{y}(t_0) = \boldsymbol{y}_0$
  - An **Initial Value Problem (IVP)** is an **ODE** with initial conditions
  - A **Solution** satisfies the **ODE** and IC on an interval containing $t_0$
  - A scalar **ODE** is **Separable** if $y' = g(t)h(y)$
- A **Linear ODE** has form $\boldsymbol{y}' = \boldsymbol{A}(t)\boldsymbol{y} + \boldsymbol{b}(t)$ with coefficients independent of $\boldsymbol{y}$
  - **Constant Coefficient (CC)** means $\boldsymbol{A}$ and $\boldsymbol{b}$ do not depend on $t$
  - The **Forcing** is $\boldsymbol{b}(t)$. **Homogeneous** means $\boldsymbol{b}(t) = \boldsymbol{0}$

## Euler's Method

- For the IVP $\boldsymbol{y}' = \boldsymbol{f}(t, \boldsymbol{y})$, $\boldsymbol{y}(t_0) = \boldsymbol{y}_0$, choose step $h > 0$

$$\boldsymbol{y}'(t_n) \approx \frac{\boldsymbol{y}_{n+1} - \boldsymbol{y}_n}{h} \quad \Longrightarrow \quad \boldsymbol{y}_{n+1} = \boldsymbol{y}_n + h\boldsymbol{f}(t_n, \boldsymbol{y}_n) \qquad t_{n+1} = t_n + h$$

- For the autonomous Rössler system $\boldsymbol{y}' = \boldsymbol{f}(\boldsymbol{y})$ with $\boldsymbol{y}(0) = (1, 0, 0)$ and $h = 0.1$

$$\boldsymbol{y}_1 = \boldsymbol{y}_0 + 0.1\boldsymbol{f}(\boldsymbol{y}_0) = (1, 0, 0) + 0.1(0, 1, 0.2) = (1.0, 0.1, 0.02)$$

$$\boldsymbol{f}(\boldsymbol{y}) = \begin{pmatrix} -y_2 - y_3 \\ y_1 + 0.2y_2 \\ 0.2 + y_3(y_1 - 5.7) \end{pmatrix}$$

| $n$ | $t_n$ | $\boldsymbol{y}_n$ |
|---|---|---|
| 0 | 0.0 | (1.0000, 0.0000, 0.0000) |
| 1 | 0.1 | (1.0000, 0.1000, 0.0200) |
| 2 | 0.2 | (0.9880, 0.2020, 0.0306) |
| 3 | 0.3 | (0.9647, 0.3048, 0.0362) |

- Practical solvers use more sophisticated methods with error control. Small steps where needed, big steps where possible!

## Scalar First-Order Linear ODE $a_1(t)y' + a_0(t)y = f(t)$

- **Standard Form** is $y'(t) + p(t)y(t) = q(t)$. In other words, just divide by $a_1(t)$.
- The aim is to muliply by $\mu(t)$ to collapse the terms on the RHS. We want

$$\mu y' + \mu p(t) y = (\mu y)' = \mu y' + \mu' y$$

- The terms match if $\mu' = p(t)\mu$. The **Integrating Factor** is the solution

$$\mu(t) = \exp\left(\int p(t)\,dt\right) \quad \Longrightarrow \quad (\mu y)' = \mu q(t)$$

- Integrate $\boxed{\mu(t)y(t) = \int \mu q\,dt + C = \mu(t_0)y(t_0) + \int_{t_0}^{t} \mu(\tau)q(\tau)\,d\tau}$
- If there is an initial condition, use it to determine $C$. If forced divide by $\mu(t)$ to get

$$\boxed{y(t) = \left(\int \mu(t)q(t)\,dt + C\right)/\mu(t)}$$

## Linear ODE Example $2y' + 4ty = 8t \quad y(0) = 3$

- Divide by 2 to get standard form

$$y' + 2ty = 4t \qquad p(t) = 2t \qquad q(t) = 4t$$

- The integrating factor is

$$\mu(t) = \exp\left(\int 2t\, dt\right) = e^{t^2}$$

- Multiply and use the product rule, then integrate

$$e^{t^2}y' + 2te^{t^2}y = 4te^{t^2} \quad \Longrightarrow \quad (e^{t^2}y)' = 4te^{t^2}$$

$$e^{t^2}y = \int 4te^{t^2}\, dt + C = 2e^{t^2} + C$$

- Use $y(0) = 3$ to get $3 = 2 + C$, so $C = 1$. Divide by $e^{t^2}$ to get $\boxed{y(t) = 2 + e^{-t^2}}$